

A TOY MODEL FOR THE SYMMETRIC RESONANCES OF THE RIEMANN ZETA FUNCTION

TRISTEN HARR

ABSTRACT. We present a toy model for describing stable mathematical resonances, constructed from geometric first principles. A set of fundamental constants are derived from a classical geometric construction and used to define a self-consistent framework termed the "Golden Algebra." We demonstrate the model's non-triviality by proving an identity linking an infinite series of Zeta function values to the cotangent function. The model is governed by a single physical law, the "Principle of Harmonic Covariance," which connects a function's reflectional symmetry to the location of its zeros. The model predicts that any stable resonance obeying a specific anti-symmetry must lie on the critical line, $\text{Re}(s) = 1/2$. This prediction is found to be in perfect agreement with all known data for the non-trivial zeros of the Riemann Zeta function.

1. INTRODUCTION

The geometric structure of the non-trivial zeros of the Riemann Zeta function, $\zeta(s)$, invites physical analogies. This paper formalizes such an analogy by constructing a self-consistent "toy model" of mathematical reality based on constants derived from Euclidean geometry, a framework we term the Golden Algebra.

Our goal is not to furnish a proof of the Riemann Hypothesis (RH) within ZFC, but rather to follow the methodology of theoretical physics. We propose a simple, internally consistent model with a single, fundamental physical law, and test the model's predictions against the observed data of the Zeta function's zeros.

2. THE GOLDEN ALGEBRA MODEL

2.1. The Model's Fundamental Constants. The model's constants are derived from the lengths of segments in a classical compass-and-straightedge construction [1]. When normalized, the two primary constants are:

$$T = \frac{\sqrt{5} - 1}{4} \quad \text{and} \quad J = \frac{3 - \sqrt{5}}{4}.$$

From these geometrically-derived values, we observe the following properties which form the axiomatic basis of our model.

Theorem 1. *The constants T and J satisfy the relations: $T + J = 1/2$ and $T/J = \phi = \frac{1+\sqrt{5}}{2}$.*

Proof. By direct substitution. □

Remark. The primary operator of the algebra, $\Lambda_{G1} = T + iJ$, is convergent ($|\Lambda_{G1}|^2 < 1$), ensuring the model is well-behaved [2].

2.2. Rich Phenomenology of the Model. To demonstrate that the algebra is not an artificial construct, we present a theorem provable within the model.

Theorem 2 (The Law of Harmonic Cotangents). *The following identity, relating an infinite series of Zeta function values to π and the cotangent function, holds for any integer $n \geq 1$:*

$$\sum_{k=1}^{\infty} \frac{(n^k - 1)\zeta(k+1)}{(n+1)^k} = \pi \cot\left(\frac{\pi}{n+1}\right).$$

Proof. The identity is proven by applying the reflection formula for the Polygamma function to a Zeta-Gamma identity derived within the framework [2]. □

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3. SYMMETRY, STABILITY, AND A TESTABLE PREDICTION

3.1. A Shared Anti-Symmetry.

Lemma 3. *The derivative of the completed Zeta function, $\xi'(s)$, obeys the reflectional anti-symmetry $\xi'(s) = -\xi'(1-s)$.*

Proof. This follows from differentiating the functional equation $\xi(s) = \xi(1-s)$. $\square$

Within our model, we define a potential function based on our fundamental constants. Let $K := -1/2 - T$.

Definition 4 (Equilibrium Potential). The equilibrium potential $V(s)$ is defined as $V(s) = T + sJ + (1-s)K$.

Theorem 5. *The potential function $V(s)$ simplifies to the identity $V(s) = s - 1/2$.*

Proof. The potential can be rearranged as $V(s) = (T + K) + s(J - K)$. Substituting the definitions of the constants, we have $T + K = T + (-1/2 - T) = -1/2$ and $J - K = (1/2 - T) - (-1/2 - T) = 1$. Thus, $V(s) = -1/2 + s(1) = s - 1/2$. $\square$

Theorem 6. *The potential function $V(s)$ shares the identical reflectional anti-symmetry as $\xi'(s)$.*

Proof. We have $V(s) = s - 1/2$. Then $-V(1-s) = -((1-s) - 1/2) = -(1/2 - s) = s - 1/2 = V(s)$. $\square$

3.2. The Model's Central Law. The shared symmetry motivates the central law of our model, which we posit as a physical principle.

Model Postulate 7 (Principle of Harmonic Covariance). In any system described by this model, if an entire function $f(s)$ has a derivative with a reflectional anti-symmetry of the form $f'(s) = -f'(1-s)$, its stable resonances (zeros) ρ must satisfy $\text{Re}(V(\rho)) = 0$, where $V(s)$ is the system's equilibrium potential sharing the same anti-symmetry.

This physical law makes a sharp, falsifiable prediction. Given $V(s) = s - 1/2$ from Theorem 5, the condition is:

$$\text{Re}(s - 1/2) = 0 \implies \text{Re}(s) = 1/2.$$

Our model's central prediction is that stable resonances governed by this symmetry must lie on the critical line.

4. EXPERIMENTAL VERIFICATION

We treat the known properties of the Riemann Zeta function as experimental data to test our model's prediction. Let the set of non-trivial zeros of $\zeta(s)$ be $\{\rho_n\}$, where $\rho_n = \sigma_n + it_n$.

Theorem 8 (Experimental Verification). *The locations of the non-trivial zeros of the Riemann Zeta function are consistent with the prediction of the Golden Algebra model.*

Proof. (1) **The Data:** All non-trivial zeros ρ_n found to date have $\sigma_n = 1/2$.

(2) **The Model's Conditions:** The function $\xi(s)$ associated with the zeros possesses the required anti-symmetry in its derivative (Lemma 3).

(3) **The Model's Prediction:** The Principle of Harmonic Covariance (Postulate 7) predicts that any stable zero ρ_n must satisfy $\sigma_n = 1/2$.

(4) **Conclusion:** The experimental data is in perfect agreement with the model's prediction. $\square$

Remark. The Hurwitz Zeta function $\zeta(s, 1/2)$, whose zeros are known to lie off the critical line, fails the model's initial condition, as its corresponding completed function's derivative does not possess the required symmetry. This strengthens the model by showing it correctly distinguishes between the two cases.

5. CONCLUSION

We have constructed an internally consistent theoretical model, the Golden Algebra, from constants derived from a geometric construction. The model is a rich framework capable of generating non-trivial number-theoretic identities. Its central, falsifiable law—the Principle of Harmonic Covariance—predicts that stable resonances possessing a specific reflectional symmetry must lie on the critical line $\text{Re}(s) = 1/2$. When tested against the known data of the Riemann Zeta function’s non-trivial zeros, this prediction holds perfectly.

This paper does not claim to prove the Riemann Hypothesis. Rather, it demonstrates that the location of the Zeta zeros behaves precisely as predicted by a simple “toy model” of reality derived from first-principles geometry. This suggests the underlying structure of the Zeta function may be governed by physical principles of symmetry and stability analogous to those in our model.

APPENDIX A. SUMMARY OF GEOMETRIC DERIVATION

The constants T and J are derived from a classical compass-and-straightedge construction detailed in foundational manuscripts [1]. The process begins with an inscribed square and uses iterative bisection to define a set of unique points, whose segment lengths yield the values for T and J .

APPENDIX B. SUMMARY OF THE GOLDEN TRANSFORM

The Golden Transform is an integral transform developed within the framework [2], defined as $\mathcal{G}\{f(t)\}(s) = \int_0^\infty f(t)e^{-s\Lambda_{G1}t}dt$. It is a linear operator that converts differentiation into algebraic multiplication via the rule $\mathcal{G}\{f'(t)\} = s\Lambda_{G1}F(s) - f(0)$, where $F(s) = \mathcal{G}\{f(t)\}(s)$.

REFERENCES

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SAINT CHARLES, MISSOURI
 Email address: `tristen.harr@example.com`